while to do so. Because of the diffusion problem, the male must wait until the female spawns, and then he must shed his sperms over the eggs. But she has had a precious few seconds in which to disappear, leaving the male in possession, and forcing him on to the horns of Trivers's dilemma. So this theory neatly explains why paternal care is common in water but rare on dry land.

Leaving fish, I now turn to the other main female strategy, the he-man strategy. In species where this policy is adopted the females, in effect, resign themselves to getting no help from the father of their children, and go all-out for good genes instead. Once again they use their weapon of withholding copulation. They refuse to mate with just any male, but exercise the utmost care and discrimination before they will allow a male to copulate with them. Some males undoubtedly do contain a larger number of good genes than other males, genes that would benefit the survival prospects of both sons and daughters. If a female can somehow detect good genes in males, using externally visible clues, she can benefit her own genes by allying them with good paternal genes. To use our analogy of the rowing crews, a female can minimize the chance that her genes will be dragged down through getting into bad company. She can try to hand-pick good crew-mates for her own genes.

The chances are that most of the females will agree with each other on which are the best males, since they all have the same information to go on. Therefore these few lucky males will do most of the copulating. This they are quite capable of doing, since all they must give to each female is some cheap sperms. This is presumably what has happened in elephant seals and in birds of paradise. The females are allowing just a few males to get away with the ideal selfish-exploitation strategy which all males aspire to, but they are making sure that only the best males are allowed this luxury.

From the point of view of a female trying to pick good genes with which to ally her own, what is she looking for? One thing she wants is evidence of ability to survive. Obviously any potential mate who is courting her has proved his ability to survive at least into adulthood, but he has not necessarily proved that he can survive much longer. Quite a good policy for a female might be to go for old men. Whatever their shortcomings, they have at least proved they can survive, and she is likely to be allying her genes with genes for longevity. However, there is no point in ensuring that her children live long lives if they do not also